

DASE7506 Project 1 Language Model BPB Technical Report

WikiText 2 model optimization and three-seed validation

Abstract: This project trains a decoder-only Transformer from random initialization under the fixed WikiText-2, BPE-2048, 256-token causal protocol. Two predictors were frozen before any test scoring. v1 is an 8-block, width-320 model with learned positional embeddings and tied input/output weights; v2 keeps the same architecture but replaces the position encoding with RoPE and selects its checkpoint on complete validation across three seeds. The frozen v2 predictor reaches 1.611178 test BPB, a 23.32% reduction against the 2.101265 course baseline and 0.034974 BPB below v1, while CPU time, peak memory and inference assets all stay inside the course limits at 3.713x, 1.939 GiB and 40.033 MiB. The evidence isolates the SwiGLU feed-forward block as the main architectural gain, rejects immediate width or depth scaling at the fixed budget, and shows that checkpoint selection on complete validation matters more than the training loss.

Version notes: this document is the complete English journal-format edition of the project record. It is not compressed to the ten-page submission limit; a separate ten-page edition is retained. The main text keeps the general conclusions while the extended pages add curves, data and decision criteria, and the appendix lists all 33 retained non-smoke experiments together with the optional figures. The ten-page requirement must still be met before submission.

1. Background and Mission Objectives

Train the language model from scratch, improve its architecture or training process, and achieve the lowest reproducible test “bits per byte” (BPB) within resource constraints. This project trains a compact decoder-only Transformer from random initialization under the course’s fixed WikiText-2, BPE-2048 tokenizer, 256-token context, and causal evaluation protocol.

The goal is to minimize the full test split bits per byte (BPB) without using external text, pre-trained weights, or test parameter tuning. For each independent window, the model predicts the next token based on previous text; Except for the first token of the split, each target scores exactly once.

Table 1. Fixed conditions for operations and rigid upper limits

Category	Requirements	How this project complies
Data	Only use train for learning; validation selection; test without parameter adjustment	Fixed split use, freeze before testing
Evaluation	BPE-2048; context 256; independent causal windows; CPU FP32	Fixed tokenizer and evaluator
CPU time	Not exceeding five times the baseline	Ultimately 2.875x for v1 and 3.713x for v2
Memory	Peak RSS does not exceed 4 GiB	Ultimately 1.948 GiB for v1 and 1.939 GiB for v2
Inference assets	Uncompressed assets must not exceed 64 MiB	Finally 40.345 MiB for v1 and 40.033 MiB for v2

Category	Requirements	How this project complies
Submission	Report no more than 10 pages; Code and checkpoint reproducible; Disclosure of AI assistance	This manuscript retains the complete material; Before submission, manually compress to 10 pages

2 Evaluation Indicators and Selection Agreement

Let B be the number of bytes of split original UTF-8 text. Fix the evaluator and accumulate negative logarithmic likelihoods for all scored targets, then convert nats to bits and divide by B .

$$BPB = \frac{-\sum_i \ln p(y_i | y < i)}{\ln(2) \times B}$$

The lower the BPB, the better. The total negative logarithmic likelihood of a complete split must be aggregated and then divided by bytes; simple averaging of BPB for batch or window is not possible. Token perplexity is only used as an auxiliary metric because course rankings are based on BPB.

Table 2. Report indicators and assessment methods

Indicators	Calculation or measurement	Purpose
BPB	Total NLL / $\ln(2)$ / UTF-8 bytes	The lower the main quality indicator, the better
Token PPL	$\exp(\text{total NLL} / \text{number of targets})$	Auxiliary explanations under fixed tokenizers
CPU time ratio	Candidate runtime / baseline runtime on the same machine	Determine if it exceeds 5 times
Peak RSS	Maximum resident memory for independent evaluation processes	Check if it exceeds 4 GiB
Inference assets	The uncompressed size of the file required for inference	Determine whether it exceeds 64 MiB

2.1 Data Usage and Freezing

Train is only used to update parameters. validation is used to select structure, learning rate, regularization, seed, and checkpoint. After selecting seed 17, the project first records SHA-256 for checkpoint, student.py, evaluate.py, and tokenizer.json, then runs the student-model test. Test results do not revert to v1's structure or hyperparameter selection.

Fair comparisons fix data, tokenizer, context, batch size, and evaluator. Short experiments uniformly process 9,830,400 training targets. When model size or training duration changes, the report provides BPB and computational cost simultaneously, avoiding miswriting extra computation as the benefits of a single mechanism.

Table 3. The scale and training budget of the main experiment

Experiment	Parameter count	Training targets	Validation BPB	Conclusion
Course baseline	1.088M	9.83M	2.071087	Reproduce the benchmark

Experiment	Parameter count	Training targets	Validation BPB	Conclusion
Small LN + GELU	1.084M	9.83M	2.101695	block comparison
Small RMSNorm + SwiGLU	1.082M	9.83M	2.040526	Smaller control: 2.91% lower
w320 d8	10.571M	9.83M	1.820432	Capacity brings a 10.79% improvement
w320 d8 Long Training	10.571M	28.91M	1.622441	Further improvement 10.88%

3 Models and Training Methods

The frozen v1 model contains 8 decoder blocks of width 320 with 4 attention heads. Each block uses pre-RMSNorm, causal scaled dot-product attention and a SwiGLU feed-forward network, and no bias is used in the linear layers. The input embedding shares its weights with the output head, positional information is supplied by learned embeddings, and dropout is 0. The model contains 10,570,560 parameters. The frozen v2 predictor keeps every block and hyperparameter of v1 but replaces the learned positional embeddings with rotary position embeddings, which changes the parameter count to 10,488,640.

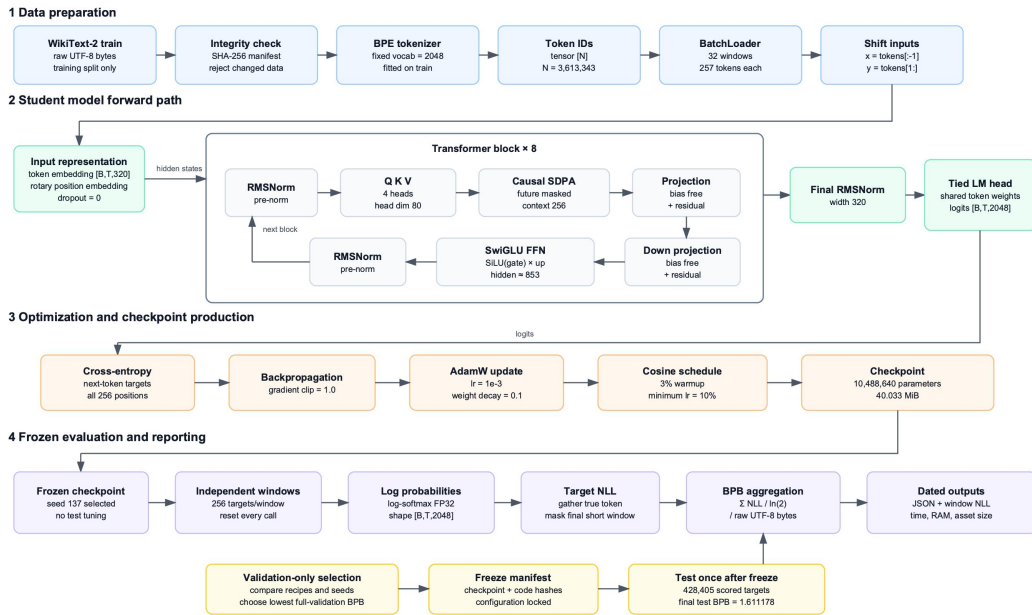


Fig. 1. Frozen v1 model structure, training data flow, and evaluation process

The training sample consists of 257 consecutive tokens, with the first 256 as input and the last 256 as targets. The batch size is 32, and each step processes 8,192 targets. The optimizer is AdamW, peak learning rate is 1e-3, weight decay is 0.1, betas are (0.9, 0.95). Learning starts with a warm-up of 3%, then drops to 10% of the peak according to cosine schedule, with gradient clipping of 1.0. Training uses Apple MPS BF16, with official scores recalculated using CPU FP32.

Each recipe and seed was independently trained from random initialization. The 28.91M target long experiment was retrained and the learning rate schedule extended,

rather than being directly continued on the 9.83M checkpoint. Frozen v1 uses the last weight; Subsequent RoPE experiments additionally add the best pre-declared checkpoint weight storage.

4 BPB Optimization Process and Judgment Methods

The development of frozen v1 is linked to the same decision path as subsequent RoPE candidates. Arrows indicate that the next round of experiments is scheduled based on the previous round results, not direct continuation; Each group is trained from random initialization

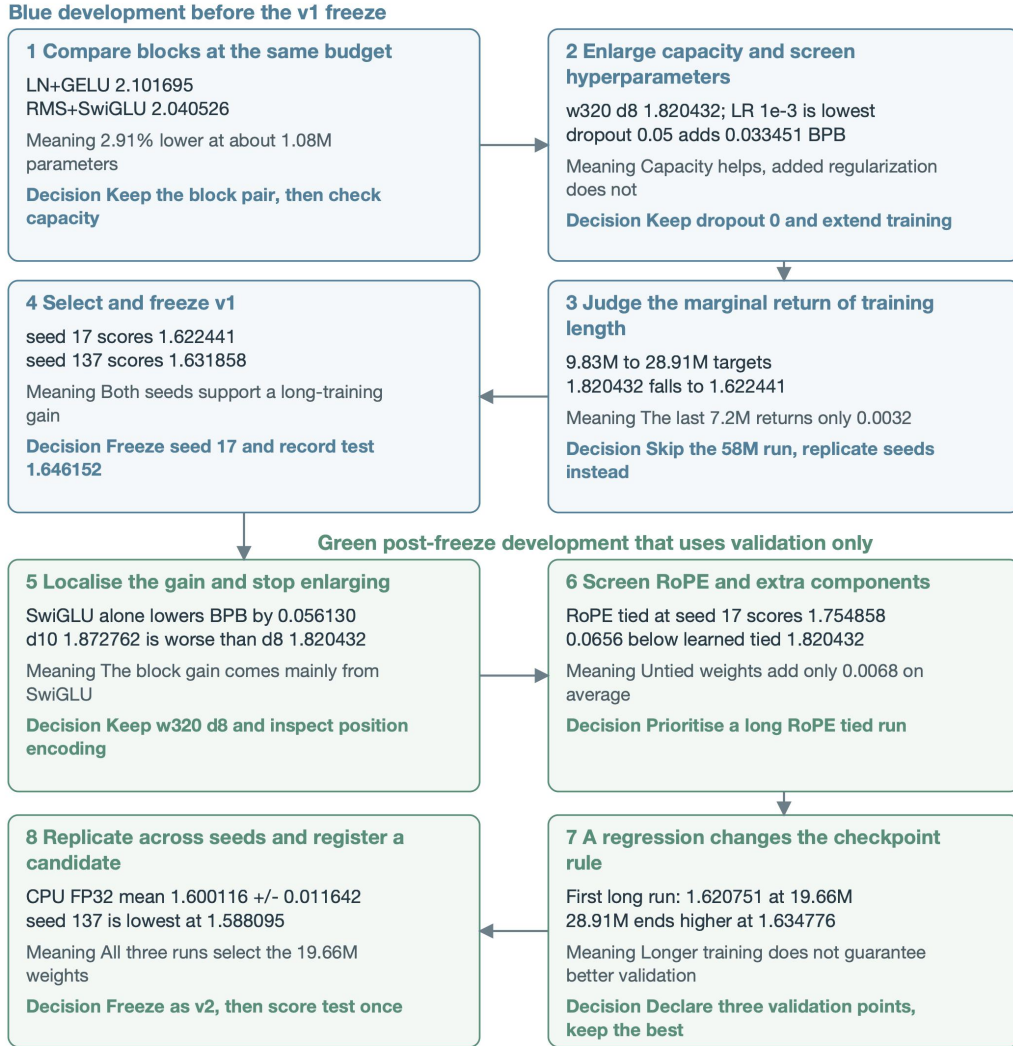


Fig. 2. Experimental data, the implications of results, and the complete process for subsequent decision-making. Boxes are read left to right and arrows mark planned decisions rather than automatic continuation; quoted values are validation BPB unless stated otherwise, short screens fix 9.83M targets, and three-seed statistics are recomputed in CPU FP32

The course baseline of 2.071087 is the starting point for reproduction and cannot replace the controlled comparison of 2.101695 shown in the figure. The combined block first achieved a 2.91% improvement, then expanded width and depth for another 10.79%,

so the budget shifted to w320 d8 hyperparameter and training length. The contribution of independent components is supplemented by ablation tests in Section 7.

4.1 Numerical decomposition of optimization paths

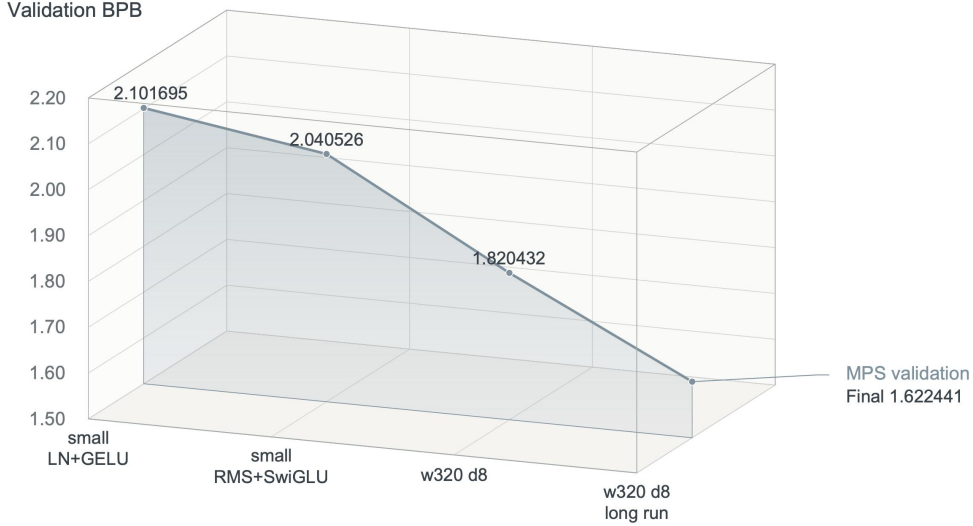


Fig. 3. Main optimization stages. Each point is an independent run; the connecting line only shows the development order. Vertical axes are truncated throughout, the translucent wall fill is not an error band, and the depth offset only separates series

The initial question is: under roughly the same scale and the same training targets, is it worth continuing to invest in block replacement? The student-side LN+GELU control is 2.101695, and RMSNorm+SwiGLU is 2.040526, a decrease of 0.061169 BPB. This comparison supports keeping the new block but cannot directly indicate how much RMSNorm and SwiGLU each contribute, so subsequent 2×2 ablation is performed. The official course baseline is 2.071087, but its training settings are different and cannot replace the controlled control here.

Then the model was scaled to w320 d8, and the short budget BPB was reduced to 1.820432. The larger gains indicate that the original small model’s capacity may have limited the current formula, so the learning rate was chosen at the new scale before deciding whether to continue scaling. Long training further dropped to 1.622441, but this step also changed the training budget and learning rate schedule, so the difference of 0.197991 could not be written as structural improvement.

Table 4. Adjacent differences at each stage and interpretation boundaries

Adjacent comparison	BPB decreases	Relatively lower	Judgement supported by the comparison
Small block replacement	0.061169	2.91%	The combination is worth continuing; A breakdown mechanism is needed
From the small model to w320 d8	0.220094	10.79%	Expansion is effective under the current short budget
w320 d8 Short to Long training	0.197991	10.88%	Increasing training investment is effective

The decision sequence narrows the problem in sequence: first confirm whether the block combination is effective, then find the appropriate size and learning rate, and finally check the marginal benefit of longer training. It is not a full factor search, nor does it exhaustively examine all combinations and seeds.

5 Basis for Decisions on Hyperparameters and Training

Length

After determining w320 d8, we first checked whether the learning rate restricts optimization. All six short experiments used seed 17 and 9.83M targets; $1e-3$ yielded the lowest BPB, while $7.5e-4$ was only 0.006146 different. $1.25e-3$, $2e-3$, and $4e-3$ were all worse, so $1e-3$ was fixed for long training and the high learning rate search was not expanded.

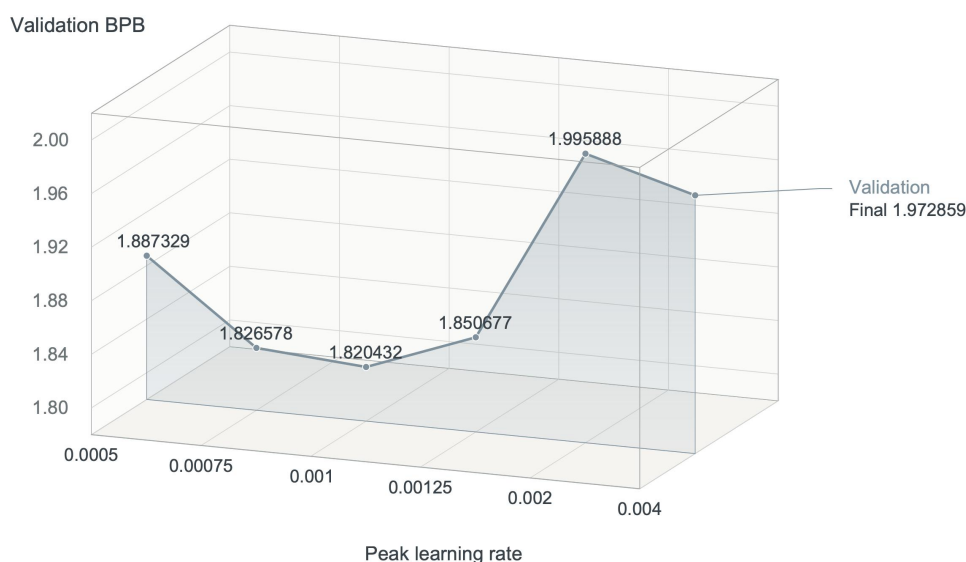


Fig. 4. Learning rate scanning with a fixed training budget

Table 5. How different outcomes change the subsequent direction

Comparison	The result	Judgment	Follow-up actions
Peak LR	$1e-3$ is 1.820432	Scan the lowest point	Fixed $1e-3$
Dropout 0.05	+0.033451 BPB	Noticeably worse	Use no dropout in v1
Weight decay 0.2	+0.000085 BPB	No meaningful gains have been observed	Keep the existing 0.1
Seed 137	+0.009417 BPB	Both support long-term training gains	Choose the lower seed 17 according to the preset rules
58M targets	Not yet running	The last improvement was about 7.2M, only about 0.0032	Don't double the training costs

Dropout 0.05 raised BPB from 1.820432 to 1.853883, while weight decay 0.2 yielded almost no benefit. This indicates that under current short training budgets, enhanced regularization is not the priority; Based on these results alone, it is not enough to

assert that the model is necessarily underfitting. We keep dropout 0 and weight decay 0.1 and instead test longer training sessions.

After expanding the full training budget from 9.83M to 28.91M, BPB dropped from 1.820432 to 1.622441, indicating that this formula can still benefit from further updates. At the end, about 7.2M targets improved from 1.625607 to 1.622441, showing a significant slowdown in return, so the 58M formula, which nearly doubled in operating costs, was not continued.

Then seed 137 is added, resulting in 1.631858. Both seeds support long training gains, but a 0.009417 difference indicates that random initialization still affects the results. We select seed 17 with the lowest validation; this provides cross-seed support but is not sufficient to accurately estimate population variance.

5.1 Why should training loss and validation be viewed together?

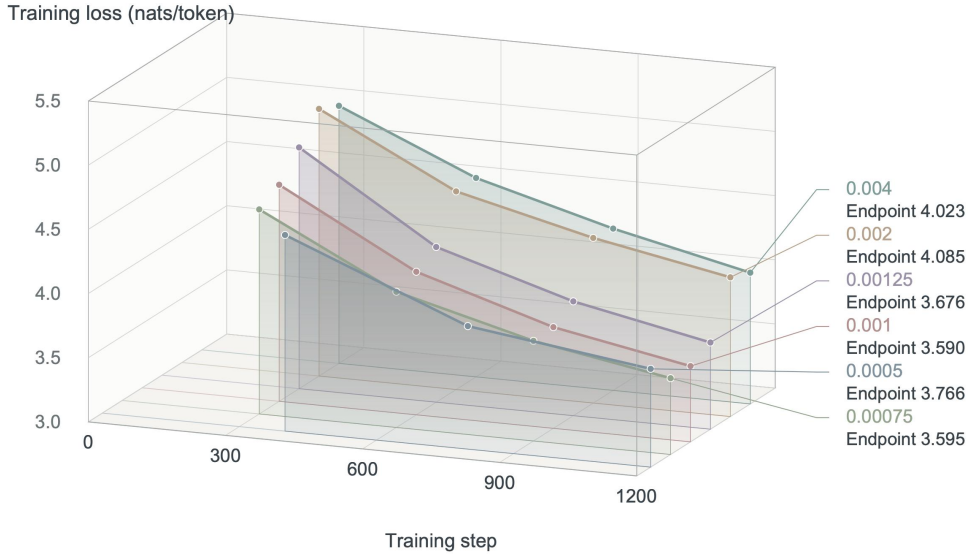


Fig. 5. Learning rate and training loss. Each point represents the batch NLL actually retained in the log; The number of recorded steps may vary depending on the run

The overall training loss for the six learning rates decreases, but these logs only record part of the batch. The 0.0005 record is at steps 400, 800, and 1200, while the other five runs are at 300, 600, 900, and 1200; In the diagram, the actual step points are placed; the first point of different runs cannot be considered the same progress. The final full validation is lowest at 0.001 at 1.820432. The record loss at 0.002 and 0.004 is higher, and the final validation is poor; under the current budget, there is no evidence to further amplify the learning rate.

Table 6. Six learning rates are recorded at the end and fully verified

Peak LR	Last item: loss	Validation BPB	An increase of 0.001	Training seconds
0.0005	3.7659	1.887329	0.066898	370.8
0.00075	3.5945	1.826578	0.006146	334.5
0.001	3.5900	1.820432	0.000000	386.7
0.00125	3.6757	1.850677	0.030245	335.2

Peak LR	Last item: loss	Validation BPB	An increase of 0.001	Training seconds
0.002	4.0850	1.995888	0.175457	403.0
0.004	4.0226	1.972859	0.152427	381.4

Therefore, the subsequent w320 d8 uses 0.001. The result for 0.00075 is 1.826578, only behind 0.006146. Existing single-seed experiments are insufficient to prove the stability gap between the two. Choosing the best observation value here is an operational decision made within a limited budget.

5.2 Scope of Application for Regular Results and Learning Rate Selection

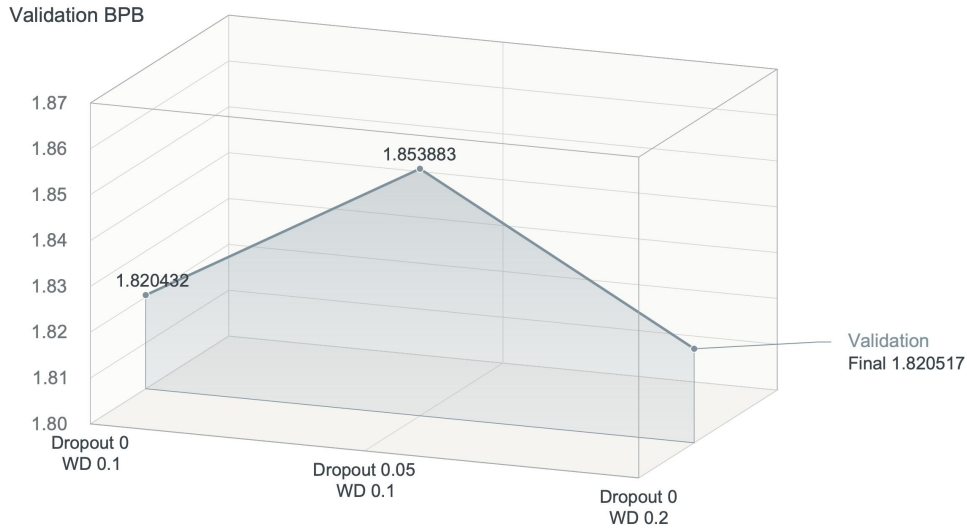


Fig. 6. Validation results for the regularization settings, comparing the same learning rate over a discrete set of configurations at a fixed training budget

Dropout 0.05 raises validation from 1.820432 to 1.853883. After increasing weight decay to 0.2, it becomes 1.820517, just 0.000085 less than the control. Current evidence does not support gaining benefits by enhancing these two regularities, so dropout 0 and weight decay 0.1 are retained. The approximate parity of a single experiment cannot be interpreted as completely equivalent to both settings.

Table 7. Results of the same hyperparameter at different scales

Scale	LR 0.001	LR 0.002	This round of judgment
Small RMSNorm + SwiGLU	2.040526	1.953693	0.002 is better
w320 d8	1.820432	1.995888	0.001 is better

The small model reached 1.953693 at LR 0.002, better than its LR 0.001 result. This record is kept because it reminds us that learning rate conclusions depend on model size and budget. After scaling the model, it must be re-filtered; the optimal settings for small models cannot be directly transferred. The mainline diagram selects the same LR 0.001 to display block and capacity comparison, rather than claiming that every small model point has been fully parameterized.

5.3 Long Training Curves and Reasons to Stop Expanding Budgets

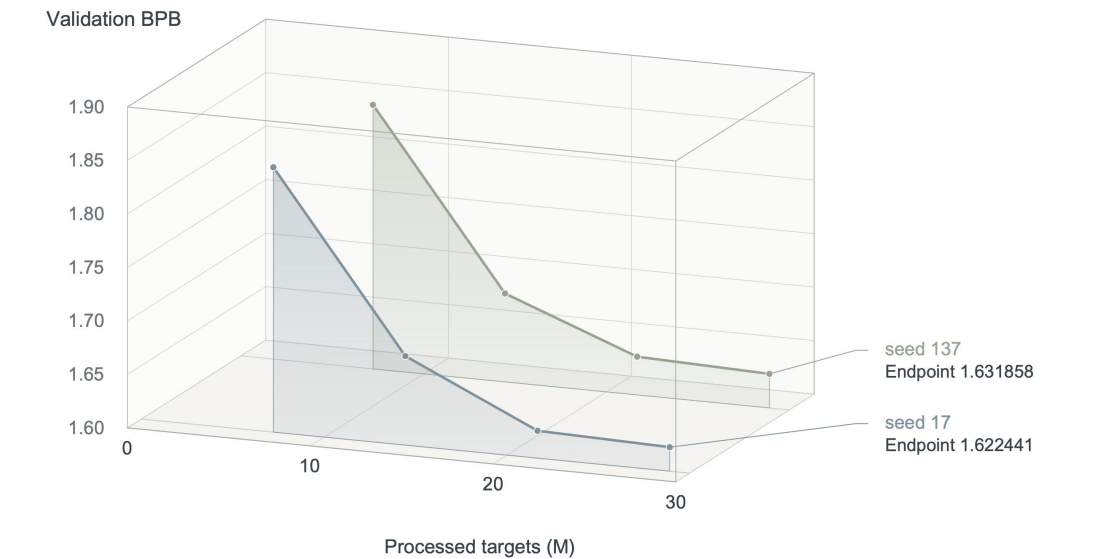


Fig. 7. Two-seed validation curves of the frozen v1 recipe. The horizontal axis is computed from the actual processed training targets; validation targets from the older logs are not used as the horizontal axis

Table 8. BPB at each validation point of the long v1 recipe

Training targets	seed 17	seed 137	Seed 17 is lower than the previous point
7. 225344M	1. 847600	1. 846897	—
14. 450688M	1. 683248	1. 682641	0. 164352
21. 684224M	1. 625607	1. 635717	0. 057641
28. 909568M	1. 622441	1. 631858	0. 003166

seed 17 dropped by about 0.164352 between the first two validation points, then by about 0.057641, and by the end only 0.003166. The marginal return was significantly reduced, so instead of immediately investing 58M targets, the second seed was used to check the recipe. The end of seed 137 was 1.631858, a difference of 0.009417 from seed 17, with both curves supporting a long recipe being better than a short one.

Finally, seed 17 is selected as validation, and only after freezing is the test evaluated. Not running 58M is just a current budget allocation decision and does not prove that longer training or rescheduling learning rates are ineffective.

6 Final Results and Resources After Freezing

Two predictors were frozen before any test scoring. v1 is the development recipe selected on validation. v2 is the seed-137 RoPE run whose checkpoint was taken at the predeclared 19.66M-target validation point and whose hashes were recorded on 26 September 2026, before the test split was scored. Each freeze records the checkpoint, implementation, evaluator and tokenizer hashes; only the frozen configuration is then scored, and the resource measurement repeats the same protocol. No test result was used to select, alter or reject either predictor.

Table 9. Results on the complete test split

Model	Test BPB	Token PPL	NLL nats	Targets	UTF-8 bytes
Course baseline	2.101265	80.8479	1,881,798.91	428,405	1,292,013
Frozen v1	1.646152	31.2239	1,474,220.05	428,405	1,292,013
Frozen v2	1.611178	29.0225	1,442,899.14	428,405	1,292,013

Against the course baseline the frozen v1 predictor is 0.455113 BPB lower, a relative reduction of 21.66%, and the frozen v2 predictor is 0.490087 BPB lower, a relative reduction of 23.32%. Both predictors score the same targets over the same raw byte count, so the differences come from probability quality rather than from a change in score coverage. The 0.034974 BPB gap between v2 and v1 is an absolute difference measured under one frozen test protocol.

Table 10. Test resource measurement of the frozen predictors

Indicator	Baseline	Frozen v1	Frozen v2	Course limit	Conclusion
CPU FP32 median time	4.209 s	12.103 s	18.619 s	No more than 5x	2.875x and 3.713x, within limit
Formal measurement range	4.209–4.238 s	12.096–12.164 s	16.357–19.695 s	One protocol per predictor	One warm-up, three repeats
Peak RSS	1.873 GiB	1.948 GiB	1.939 GiB	4 GiB	Within limit
Inference assets	4.168 MiB	40.345 MiB	40.033 MiB	64 MiB	Within limit

6.1 Training and Search Costs

The final V1 seed 17 training time was 1,143 s, and seed 137 was 1,093 s. Currently, the 33 non-smoke runs retained in the metrics total 15,530 s, about 4.31 hours. This statistic includes supplementary experiments after freezing, summarized by train_seconds fields for each run; It does not include separate CPU reviews, resource measurements, data preparation, or document generation.

The three long RoPE training sessions recorded approximately 1,436, 1,399, and 1,349 seconds respectively. Although the final weight came from 19.66M targets, all three searches actually ran up to 28.91M, so the full search volume was used when disclosing costs. Training time comes from single-machine operation records, which is suitable for indicating project investment and should not be directly regarded as hardware-independent costs for different mechanisms.

6.2 What does formal testing correspond to?

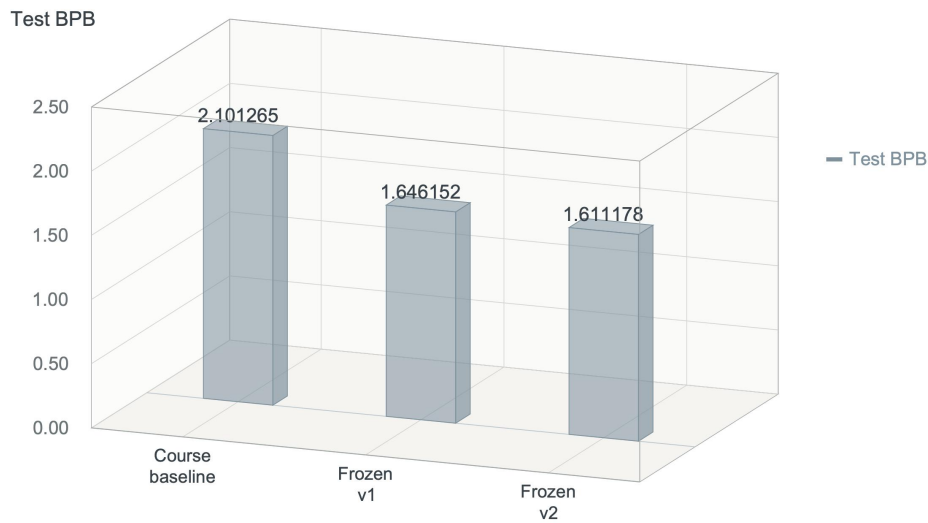


Fig. 8. Official test results of the frozen predictors. Bars start from zero so that the absolute BPB differences stay readable

Table 11. Total test volume and auxiliary indicators

Item	Course baseline	Frozen v1	Frozen v2
BPB	2.101265	1.646152	1.611178
Token PPL	80.848	31.224	29.023
Total NLL (nats)	1881798.906	1474220.050	1442899.139
Rated targets	428405	428405	428405
Original UTF-8 bytes	1292013	1292013	1292013

All three models score the same test text over the same number of targets with the same tokenizer, so the BPB differences correspond to differences in total predictive negative log-likelihood on identical text. The v2 predictor improves on the course baseline by $(2.101265 - 1.611178) / 2.101265$, about 23.32%, and on v1 by 0.034974 BPB. Token perplexity falls in step, but the ranking and the main conclusions are still expressed in BPB.

This section reports the two frozen predictors only. The validation score of 1.588095 that selected v2 belongs to the validation split and must not be subtracted from a test score; it is quoted only to document the selection rule that produced the v2 checkpoint.

6.3 Resource thresholds and measurement protocols

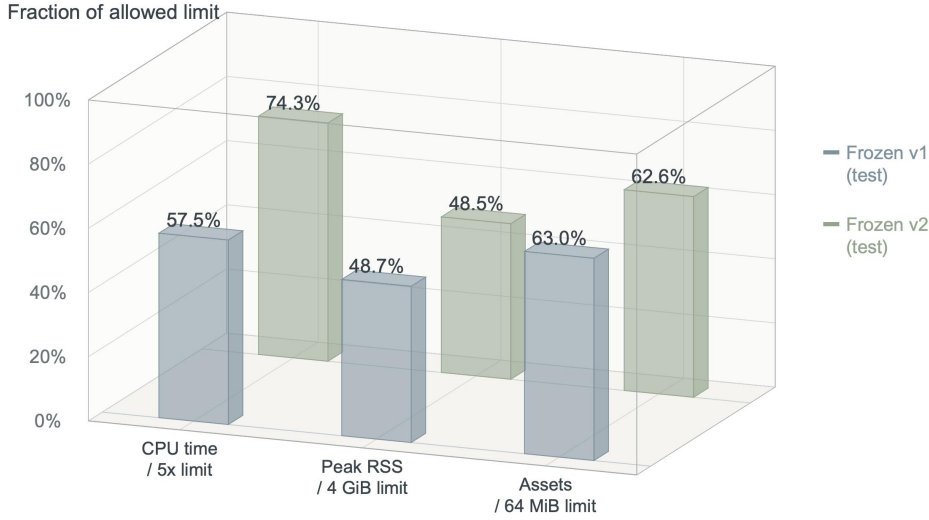


Fig. 9. Resource use against the course limits. Both frozen predictors stay below their respective limits; the standardised ratios are used only to assess compliance

Table 12. Resource records of the frozen predictors

Metric	Frozen v1 (test)	Frozen v2 (test)
Median CPU time (seconds)	12.103	18.619
Range over three runs (seconds)	12.096–12.164	16.357–19.695
Median baseline of the same protocol (seconds)	4.209	5.014
CPU time ratio / limit 5x	2.875x	3.713x
Peak RSS / limit 4 GiB	1.948	1.939
Inference assets / limit 64 MiB	40.345	40.033

Measurements use CPU FP32 with four threads and three independent processes after one warm-up. Median time is reported and peak RSS is the largest value observed across processes. Both frozen predictors are measured on the test split under the same protocol on the same machine, so the time ratios share one baseline and the durations may be compared directly.

A lower BPB is only a valid entry when the time, memory and asset limits are met. The wider 384 model stays inside the 64 MiB asset limit, but its quality at the same budget is worse than w320 d8, so it was not carried into long training.

7. How the supplementary ablation changes the subsequent direction

The following experiment was completed after the v1 test, explaining the mechanism only with validation and filtering subsequent candidates, corresponding to the lower half of Figure 2. The short experiment fixed 9.83M targets, with values from MPS records. When the additional gain is less than 0.01 BPB, long-term training is not prioritized;

This empirical threshold is used to control search cost and is not equivalent to statistical significance testing.

Table 13. Normalization and FFN Ablation at the Same Budget

Normalization	FFN	Validation BPB	Compared to LN+GELU
LayerNorm	GELU	2.101695	0
RMSNorm	GELU	2.095283	-0.006411
LayerNorm	SwiGLU	2.045564	-0.056130
RMSNorm	SwiGLU	2.040526	-0.061169

With fixed LayerNorm, SwiGLU improved by 0.056130 BPB; When GELU was fixed, RMSNorm improved by only 0.006411. Therefore, the main yield of the block portfolio is supported by SwiGLU, and RMSNorm’s independent yield remains small. This group uses seed 17 and cannot write single differences as proven cross-seed stability effects.

Table 14. Supplementing experiments with observation and improving decision-making

Problem	Experimental observation	Judgment and subsequent actions
Does the benefit come from the trainer?	Baseline 2.072376; no return 2.071598; then beta2 changed to 2.103114; full formula 2.098476	Sampling is nearly neutral, and beta2 deteriorates on a small baseline. The trainer does not separately explain structural benefits and continues to check blocks.
Can the capacity be further increased?	w320 d8 is 1.820432; D10 is 1.872762; W384 D8 is 1.829156	The larger model did not win at the same budget; The latter had assets of 57.422 MiB. Keeping the w320 d8, shifting the budget to the presentation layer.
Position and weight sharing	Seed 17: learned tied 1.820432, untied 1.786678, and RoPE tied 1.754858; the two-seed mean for RoPE was 1.748021	RoPE prioritizes long training. Afterwards, paired pairs with untied have an average additional gain of about 0.00675, below the threshold, so tied is still used.
Is it worth holding on to the EMA?	For the same workout, live is 1.735489, EMA is 1.734143, and the improvement is 0.001346	Only compare the same optimization trajectory, avoiding calculating different run fluctuations as EMA gains. Improvements are too small, without advancing long training.
Is MTP worth keeping?	MTP is 1.739964; the control RoPE untied seed 17 is 1.745039, an improvement of 0.005075	Improvement less than 0.01, not advanced. Single run time is insufficient to confirm fixed cost increases, requiring controlled retesting.

7.1 First, exclude the trainer itself that causes the main benefits

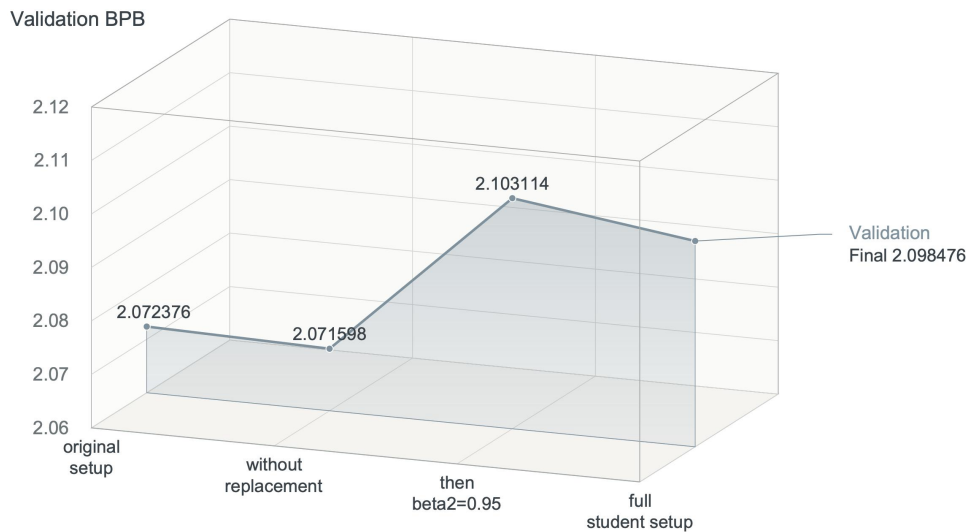


Fig. 10. Trainer settings for comparison. Four points are arranged in the order of change in the settings; All are small models, seed 17, and 9.83M targets

Table 15. The trainer sets incremental changes

Setup	Validation BPB	Relative changes in the previous item
Reproduce the baseline trainer	2.072376	—
Without-replacement sampling	2.071598	-0.000778
Then change beta2 to 0.95	2.103114	0.031516
Complete student training setup	2.098476	-0.004638

Without-replacement sampling changed from 2.072376 to 2.071598, improving by only about 0.000778, nearly flat. Beta2 was then changed to 0.95, BPB rose to 2.103114; full student training was set to 2.098476, also not better than the starting point. Therefore, the main benefit of v1 cannot be attributed to sampling method or trainer replacement.

Next, focus on the structure: keep the student-side trainer and replace norm and FFN respectively. Here, 2.098476 is the trainer reference for the course small model structure, and 2.101695 for the next page's student-side LN+GELU comes from different settings and cannot be mixed.

7.2 Splitting RMSNorm's Contributions to SwiGLU

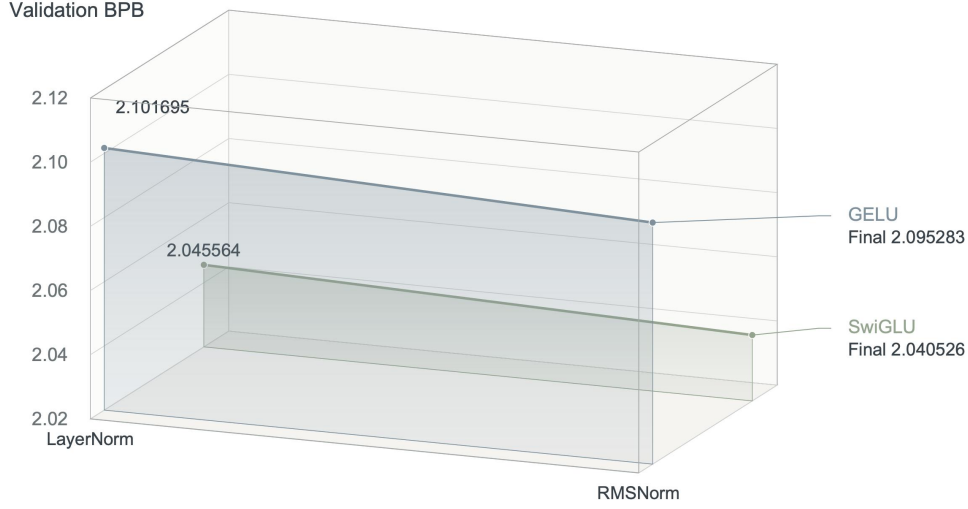


Fig. 11. 2×2 ablation of norms and FFNs. Two norms and two FFNs form a 2×2 control, all under the same short budget

Table 16. Relative improvement in LN+GELU

Setup	Validation BPB	BPB decreases
LN+GELU	2.101695	0.000000
RMS+GELU	2.095283	0.006411
LN+SwiGLU	2.045564	0.056130
RMS+SwiGLU	2.040526	0.061169

SwiGLU alone brought about 0.056130 BPB improvement, and RMSNorm alone brought about 0.006411. The combination improvement was 0.061169, so this round of block earnings is mainly related to SwiGLU replacement. The two independent differences are not strictly summed, so the combined results cannot be written as completely independent mechanisms.

Accordingly, RMSNorm+SwiGLU is retained as the subsequent infrastructure, but no longer uses the single norm replacement as the main search direction. Currently, this attribution only has evidence for seed 17, and it cannot yet be proven that the difference remains exactly the same after seed change.

7.3 Why Deeper, Wider Didn't Continue

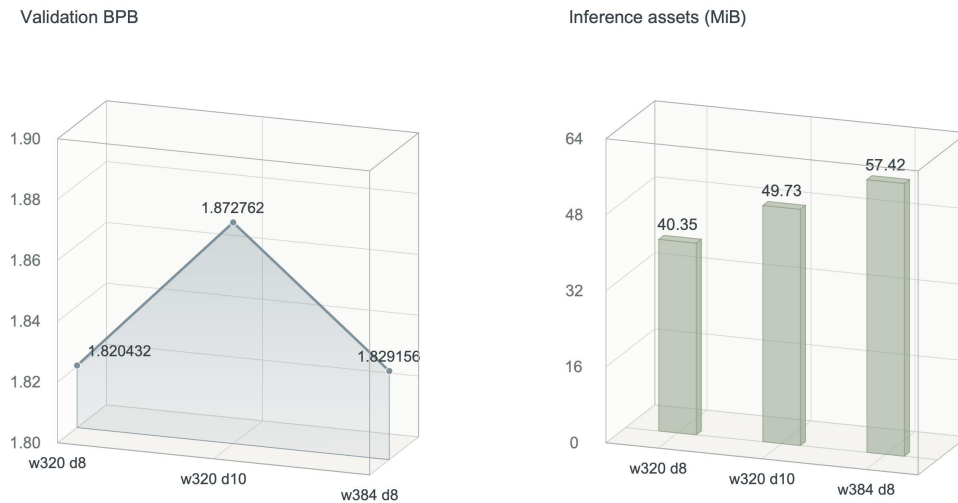


Fig. 12. Short-budget comparisons for deeper or wider models. Quality and assets are displayed separately, avoiding placing both units on the same vertical axis

Table 17. Capacity, quality, and training costs

Setup	Parameter value M	BPB	Assets MiB	Training seconds
w320 d8	10.571	1.820432	40.345	386.7
W320 D10	13.029	1.872762	49.728	490.6
w384 d8	15.047	1.829156	57.422	459.7

After increasing to layer 10, BPB is 1.872762, which is 0.052330 higher than d8; after expanding to 384, it is 1.829156, which is 0.008725 higher than the original width. Both parameters and asset and recorded training time have increased, but the quality of the current short budget has not improved, so we will continue to use the w320 d8 going forward.

This result only negates the priority of "immediate scaling with the same targets and hyperparameters in this round." Larger models may require different learning rates or longer training, and existing experiments do not cover these combinations. To avoid allocating search budgets to directions where returns have not yet been shown, the next step is to choose to change position representation.

7.4 Short experiments with position representation and output weights

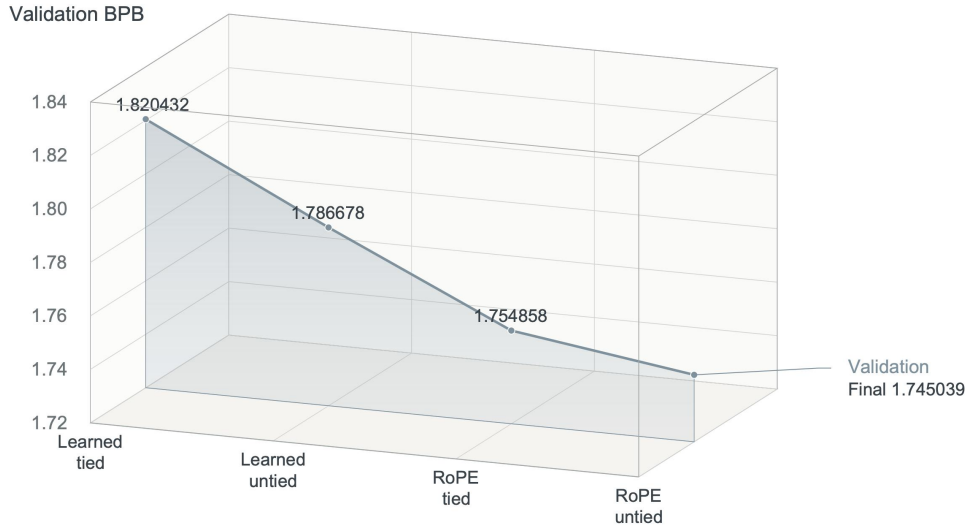


Fig. 13. Positional representation and weight sharing. All four points are seed 17; Connecting lines with different settings does not indicate continuous training

Table 18. Comparison of position and weight sharing within the same seed

Setup	BPB	Reduction relative to learned tied	Assets MiB
Learned / tied	1.820432	0.000000	40.345
Learned / untied	1.786678	0.033754	42.846
RoPE / tied	1.754858	0.065574	40.033
RoPE / untied	1.745039	0.075393	42.533

Replacing learned position with RoPE while continuing to share input/output weights lowers BPB from 1.820432 to 1.754858, an improvement of 0.065574. Separate unsharing also improves but requires a larger output weight matrix. RoPE-tied assets are about 40.033 MiB, slightly lower than learned tied's 40.345 MiB, so RoPE is first considered a long-term training candidate.

Changing position representation and weight sharing at the same time mix two factors; you can't claim all benefits come from untied sharing just by looking at the lowest RoPE untied point. Next, under RoPE conditions, pair two identical seeds to check for additional untied contributions.

7.5 Is the average additional gain from untied weights worth the investment?

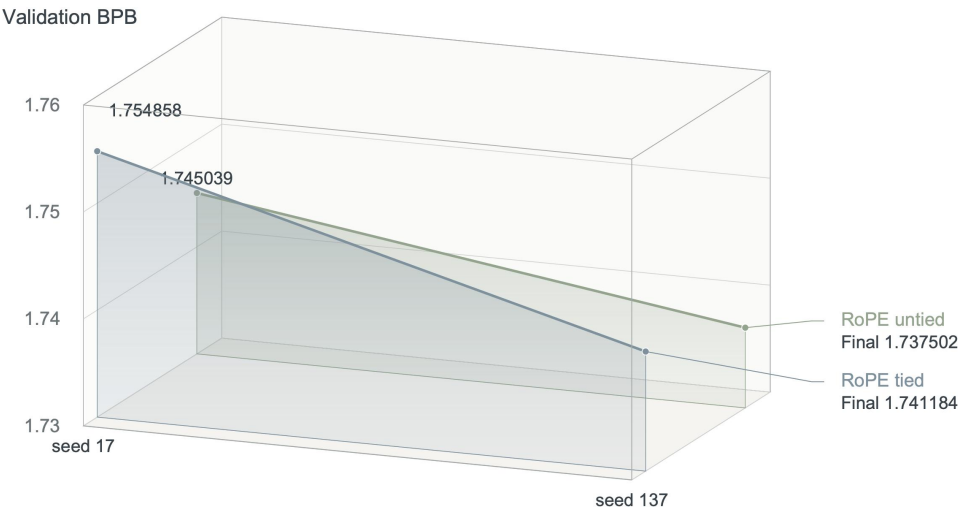


Fig. 14. Two-seed gain from untied weights under RoPE. Within the same seed, compare tied and untied, then sum up the differences

Table 19. Paired comparison of tied and untied RoPE models

Seed	Tied BPB	Untied BPB	Further reduction
17	1.754858	1.745039	0.009819
137	1.741184	1.737502	0.003682
Mean	1.748021	1.741270	0.006750

Both seeds had better untied, with an average additional improvement of 0.006750 BPB, while assets increased from about 40.033 to 42.533 MiB. This improvement was not zero, but was below the 0.01 BPB threshold used for long-term training budget allocation in this project. Therefore, priority was given to RoPE tied with simpler long training structures and smaller assets.

0.01 is the engineering input threshold, not a statistically significant standard; Two seeds are not enough to yield a stable overall effect. The untied result is kept here for future review when the budget allows, rather than being written as a failure mechanism.

7.6 Reference basis for EMA and MTP

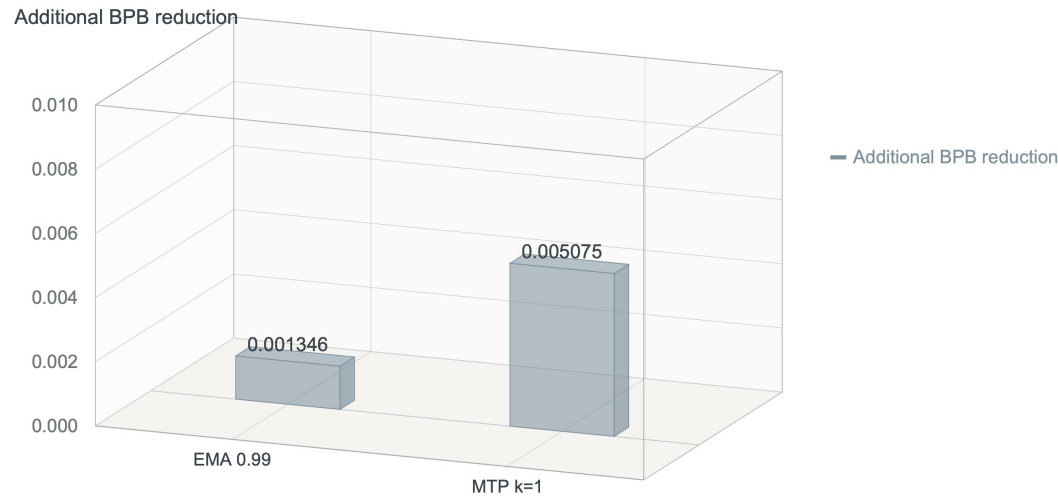


Fig. 15. Additional gains from EMA versus MTP. The two columns use different but matching controls and should not be directly ranked as the advantages or disadvantages of general methods

Table 20. Actual comparison of additional components

Components	Compare with BPB	After treatment with BPB	Further reduction
EMA 0.99 versus the live weights from the same run	1.735489	1.734143	0.001346
MTP k=1 with the matched untied recipe	1.745039	1.739964	0.005075

EMA needs to be compared with the live weight in the same run, so the return corresponds to the average operation itself. For the same run, live is 1.735489, averaged is 1.734143, with an actual difference of about 0.001346; You can't compare to another unified run at 1.745039; attribute all differences between runs to EMA.

MTP is 1.739964, which is about 0.005075 improvement compared to matched untied, also below the 0.01 threshold. Therefore, this round did not continue to allocate a long training budget to EMA or MTP. The number of training seconds per log is affected by run order and system load, so it is not appropriate to write a fixed percentage increase in training overhead for MTP.

8 RoPE Across Three Seeds and Best-Checkpoint Selection

The first RoPE long training reached 1.620751 targets at 19.66M, but rebounded to 1.634776 at the end, with only the final weight saved. This observation directly changed the training process: before subsequent reruns, three validation points—9.83M, 19.66M, and 28.91M—were declared, and the lowest validation BPB weights were saved.

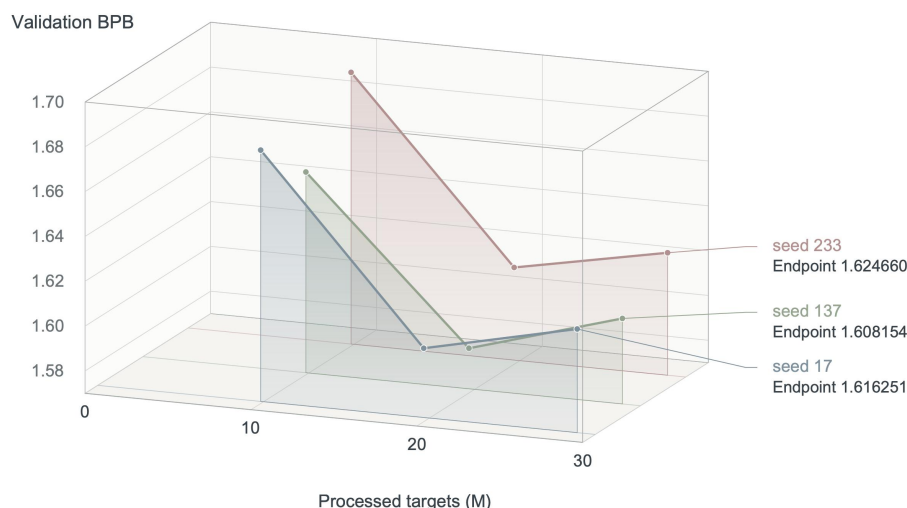


Fig. 16. Validation curves and selection locations for the three RoPE seeds

Table 21. Best-checkpoint CPU FP32 review against the end of training

Seed	Search targets	Selected targets	CPU FP32 BPB	End MPS BPB
17	28.91M	19.66M	1.600916	1.616251
137	28.91M	19.66M	1.588095	1.608154
233	28.91M	19.66M	1.611337	1.624660

The mean CPU FP32 validation for the three seeds is 1.600116, with a sample standard deviation of 0.011642. The weight of step 2400 is selected all times, but the search continues to 28.91M. The validation curve supports saving the best checkpoint instead of using the final weight.

Under the MPS caliber during training, the mean RoPE for seeds 17 and 137 was 1.594506, v1 was 1.627150, an improvement of 0.032644. Here, only the two seed subsets are compared; RoPE also changes checkpoint selection, so the entire difference cannot be attributed to position coding.

Table 22. Test resource check of the frozen v2 predictor

Indicator	Measured on test	Course limit	Outcome
CPU FP32 time ratio	3.713x baseline	5x	Within limit
Peak RSS	1.939 GiB	4 GiB	Within limit
Inference assets, uncompressed	40.033 MiB	64 MiB	Within limit

seed 137 was registered as a candidate on the lowest CPU FP32 validation score, 1.588095 BPB. Resource measurement used the validation split, four threads, one warm-up and three independent process repeats, and the checkpoint hash was recorded. The candidate has since been frozen as v2 and scored once on the test split; the freeze manifest records the hashes and the pre-test freeze timestamp.

8.1 How the first long run changed the weight-saving rule

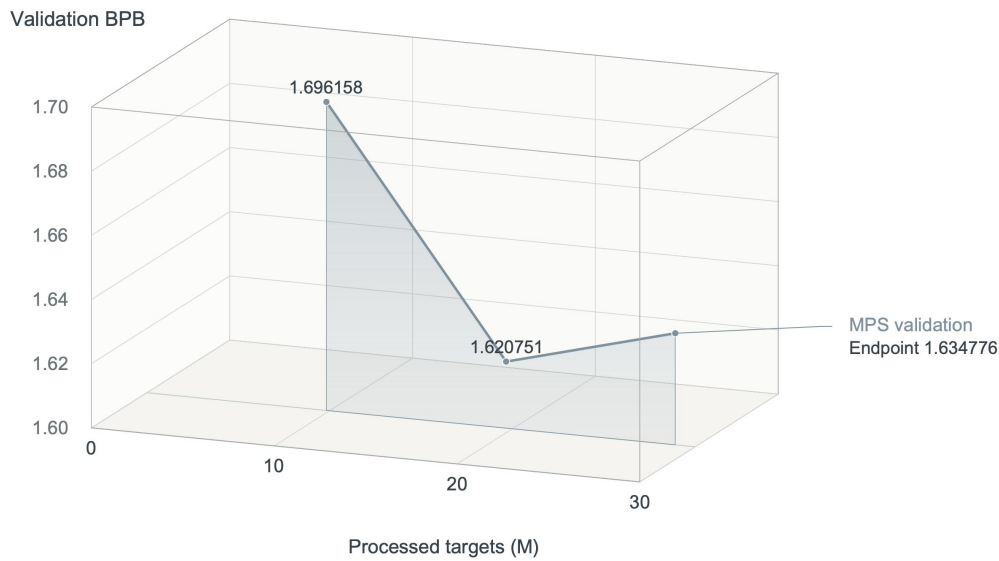


Fig. 17. Late regression of the initial long RoPE run. The initial long experiment and subsequent best-checkpoint reruns are recorded separately and cannot be combined into the same run

Table 23. Complete validation points for the initial long RoPE training

Training targets	MPS validation BPB	Weight retention status
9.830400M	1.696158	The score is retained, but the weight of the point is not retained
19.660800M	1.620751	The score is retained, but the weight of the point is not retained
28.909568M	1.634776	Only the last weight is preserved

The initial long training reached 1.620751 near 19.66M targets, but the last 28.91M rose to 1.634776, a setback of about 0.014025. Since only the last checkpoint was saved at that time, the intermediate validation score could not be used as a submittable model score. To validate and retain usable weights, new independent training was conducted afterward.

The new run pre-declares three complete validation points: 9.83M, 19.66M, and 28.91M. Each improvement saves the corresponding weight, and the training still completes the entire search budget. This separates the "actual training investment" from "where the final selected weights are trained," avoiding mistakenly writing the 28.91M search cost as 19.66M.

8.2 Why a Lower Training Loss Does Not Justify Selecting the Final Checkpoint

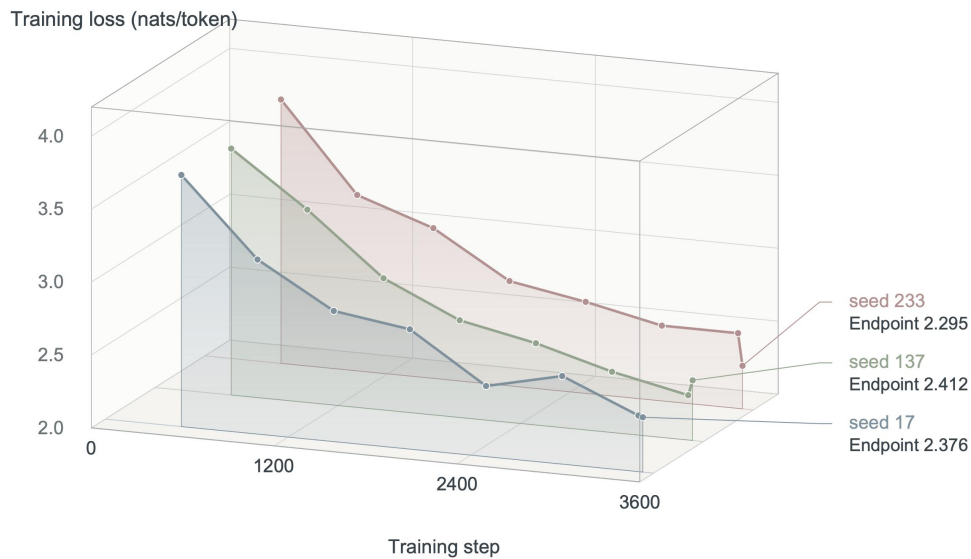


Fig. 18. Training loss of the three RoPE seeds. The connection line only connects the real log points, while the points of adjacent steps maintain actual distance

The overall training loss for the three RoPE seeds decreased from about 3.7–3.8 to about 2.3–2.4, but all validations rebounded after 19.66M. Training objectives continued to be fitted and did not guarantee continued improvement in predictions for retained text. This separation is compatible with overfitting or mismatches between learning rate and training length; Existing experiments do not distinguish between the two causes, so the report does not attribute regression to the single mechanism.

Table 24. Three new operational weights and real input

Seed	Select the step	Weight targets	Search for targets	Training seconds
17	2400	19.66M	28.91M	1435.9
137	2400	19.66M	28.91M	1399.2
233	2400	19.66M	28.91M	1348.9

Next, validation is used to select the optimal weight, and the CPU is rescored at FP32, rather than selecting the seed based on the last training loss. All three reruns selected the same budget position, indicating that this rule adjustment actually affects all three observed seeds; This does not guarantee that the optimal position remains at 19.66M after budget change.

8.3 Pairing long schemes and comparison with candidate boundaries

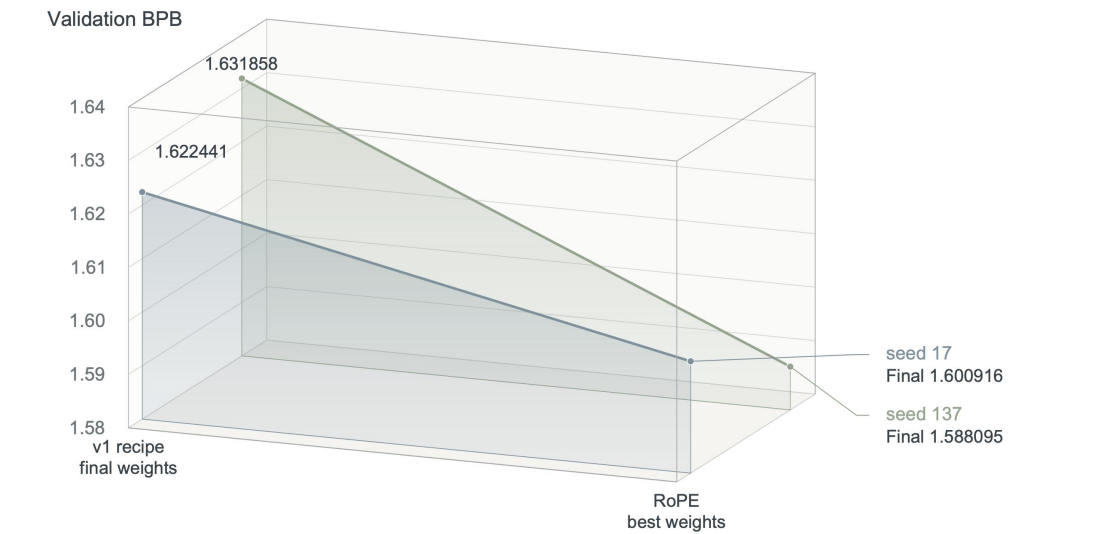


Fig. 19. v1 paired against the long RoPE recipe. Only common seeds 17 and 137; seeds 233 were additionally counted in the RoPE three-seed statistics

Table 25. The long-scheme difference of the common seed (all in MPS).

Seed	v1 Ends with BPB	RoPE best BPB	BPB decreases
17	1.622441	1.600916	0.021525
137	1.631858	1.588095	0.043763
The mean of both seeds	1.627150	1.594506	0.032644

Among the two common seeds the long RoPE recipe averages about 0.032644 BPB below v1 in MPS validation. After adding seed 233 the mean across the three RoPE seeds is 1.600116 with a sample standard deviation of 0.011642. The paired two-seed mean and the three-seed summary answer different questions and must not be mixed.

The CPU FP32 validation of seed 137 is 1.588095, so the run was registered as the v2 candidate, frozen with recorded hashes and then scored once on the test split at 1.611178 BPB. The paired difference also includes the change in positional representation and the change in the checkpoint selection rule, so it cannot be read as an isolated contribution of RoPE.

9 Correctness and submission requirements

Course fixed code, data, and tokenizer are checked via manifest hashes. Achieves non-caching validation/test answers, no access to future tokens, no cross-window state retention, and no networking required for evaluation. Contract testing covers causality, probability normalization, sample independence, window reset, misaligned target gradient, and short window at the end; Trainer tests check sampling, learning rate, and optimal checkpoint retention. RoPE broadcast maintenance curve horizontal axis repair has been documented in the experiment document.

9.1 Reproducing Materials

The v1 freeze checkpoint is at code/runs/a_e8_lr1e3/checkpoint.pt with SHA-256 ab9ba8b648513530c528ce00345ba1dd8749351c8cf5516dc5ace84f40ad6657. The v2 freeze

checkpoint is at `code/runs/2026-09-25_rope_long_best_seed137/checkpoint.pt` with SHA-256 723b8f6fe2edef1e01987a29ccf2d195945961b12dd37b0fcb5f24c9fff24e68. Validation scores, resource measurements, seed comparisons and freeze manifests are stored under `code/results/2026-09-23/` and `code/results/2026-09-26/`. The environment is Python 3.13.12, PyTorch 2.7.1, NumPy 2.5.3 and tokenizers 0.21.4; Python 3.12 is still recommended for the release environment.

9.2 Assignment Requirement Coverage

Table 26. Course requirements and evidence locations

Course requirements	Evidence in this report or the repository	Status
From random initialization training, only the course train is used	Sections 1 and 3; README data rules	Done
Fixed BPE-2048, context 256, and evaluator	Sections 1 and 2; Hash check	Done
Initial baseline	Table 3 and Table 5	Done
Comparison at the same number of processed targets	Table 3 shows a 9.83M comparison	Done
Ablation of the key mechanism	The 2×2 control in Table 7 and the screening in Table 8	Done
Validation freezes before development and testing	Sections 2 and 6; freeze manifests for v1 and v2	Done
Quality and computational cost analysis	Table 3 and Table 6 with reasons for stopping 58M	Done
Training cost, seeds, and checkpoint ancestry	Sections 3, 5, 6, and 8; grid.tsv	Done
5x CPU, 4 GiB, and 64 MiB limits	Table 6	Within limit
A report no more than 10 pages long with AI disclosure	Section 10 has disclosed AI; This material draft needs to be abridged	Compression required
Immutable code links and checkpoint download links	The last course submission must be submitted before the deadline	Pending external submission
Course website registration and subsequent peer review	Register 29 days before the date; Resubmit another copy after publication	Pending external submission

By September 29, you must register your student ID and full-test BPB and complete the generated GitHub issue; Final submission on September 30 must include immutable code and a matching checkpoint download link, providing precise installation, training, and evaluation commands. Within 7 days after publication, reproduce another submission according to the course process and report the score. The repository README has disclosed AI help; website submissions and external link status must be verified at submission.

10 Conclusions, Limitations, and AI Disclosure

10.1 Conclusion

Two predictors were frozen before any test scoring. Frozen v1 lowers test BPB from 2.101265 to 1.646152, and frozen v2 lowers it further to 1.611178, a 23.32% reduction against the course baseline with all three resource indicators inside their limits. The experiment first supports keeping the efficient block pair and scaling capacity, then fixes the learning rate, the regularization settings and the training length. Additional controls attribute most of the block gain to SwiGLU, and the position-encoding experiments combined with the late-training regression produced both RoPE and the best-checkpoint rule.

The three RoPE seeds reach a mean CPU FP32 validation of 1.600116 with a sample standard deviation of 0.011642; seed 137 is lowest at 1.588095 and sits inside the resource thresholds. The frozen v2 predictor converts that candidate quality into a test score of 1.611178 BPB. Negative results also shaped the path: deeper and wider models were not carried forward, and untied weights, EMA and MTP were stopped because their returns stayed below the 0.01 BPB investment threshold.

10.2 Limitations

v1 only compared two seeds, RoPE long training compared three, and most other short ablations only had 17 seeds. There is still uncertainty between mean and slight differences; 0.01 BPB is just the search decision threshold. The RoPE long scheme also changes the checkpoint selection method, so the benefits of long training cannot be fully attributed to position coding. No 58M experiment was run, and it cannot be ruled out that longer training still yields benefits after recalibration.

Training runs on MPS BF16 while official scores use CPU FP32, and the short screens are not all re-verified on CPU. Resources are measured on one machine, so absolute times depend on hardware and system load. Both frozen predictors are now measured on the test split under a single protocol, which removes the earlier mismatch between a validation resource check and the test submission.

10.3 AI assistance disclosure

OpenAI Codex assisted with code diagnosis, experiment planning, implementation review, script fixing, result organization, charts, and document layout. All training and evaluation values were generated by the project's scripts and retained machine-readable evidence. Codex did not provide external training text, pre-training weights, validation answers, or test answers. Submitters are responsible for understanding implementations, verifying evidence, and completing the final submission.

References

- [1] DASE7506 Mini Project 1 Guide. GUIDE.md.
- [2] DASE7506 MP1 Starter Code README. code/README.md.
- [3] Vaswani et al. Attention Is All You Need. NeurIPS 2017.
- [4] Zhang and Sennrich. Root Mean Square Layer Normalization. NeurIPS 2019.
- [5] Shazeer. GLU Variants Improve Transformer. arXiv:2002.05202, 2020.

- [6] Su et al. RoFormer: Enhanced Transformer with Rotary Position Embedding. arXiv:2104.09864, 2021.
- [7] Loshchilov and Hutter. Decoupled Weight Decay Regularization. ICLR 2019.

Appendix A preserves all experiments and source indexes

This appendix covers 33 non-smoke runs, train_seconds total 15,530.3 seconds (4.314 hours), excluding verification and debugging time. Each group is initialized independently. BPB transcript metrics.json validation, RoPE best is the MPS selection value, CPU FP32 values are shown in Table 9. EMA behavior average weight; Budget column represents actual search volume.

A.1 Baseline, Learning Rate, and Regularization

Table 27. Retained run indicators

Numbering and setup	Seed	Targets M	Val. BPB	Training seconds	Parameter M	Assets MiB
01 Course Baseline	17	9.83	2.071087	186.4	1.088	4.168
02 Small LN+GELU	17	9.83	2.101695	85.8	1.084	4.147
03 Small RMSNorm+SwiGLU / LR .001	17	9.83	2.040526	93.9	1.082	4.139
04 Small RMSNorm+SwiGLU / LR .002	17	9.83	1.953693	93.4	1.082	4.139
05 w320 d8 / LR .0005	17	9.83	1.887329	370.8	10.571	40.345
06 w320 d8 / LR .00075	17	9.83	1.826578	334.5	10.571	40.345
07 w320 d8 / LR .001	17	9.83	1.820432	386.7	10.571	40.345
08 w320 d8 / LR .00125	17	9.83	1.850677	335.2	10.571	40.345
09 w320 d8 / LR .002	17	9.83	1.995888	403.0	10.571	40.345
10 w320 d8 / LR .004	17	9.83	1.972859	381.4	10.571	40.345
11 Dropout .05	17	9.83	1.853883	343.2	10.571	40.345
12 Weight decay .2	17	9.83	1.820517	366.9	10.571	40.345

Corresponding run directory (located at code/runs/)

- 01 R0_baseline
- 02 s0_lr1e3
- 03 s_lr1e3
- 04 s_lr2e3
- 05 lr5e4_a
- 06 lr7p5e4_a
- 07 lr1e3_a
- 08 lr1p25e3_a
- 09 lr2e3_a
- 10 lr4e3_a
- 11 drop005_lr1e3_a
- 12 wd2e1_lr1e3_a

A.2 Trainer, block, and capacity ablation

Table 28. Retained run indicators

Numbering and setup	Seed	Targets M	Val. BPB	Training seconds	Parameter M	Assets MiB
13 Original baseline setup	17	9.83	2.072376	77.0	1.088	4.168
14 No return sampling	17	9.83	2.071598	75.8	1.088	4.168
15 No replacements + beta2 .95	17	9.83	2.103114	73.6	1.088	4.168
16 Complete student training setup	17	9.83	2.098476	77.4	1.088	4.168
17 RMS+GEL	17	9.83	2.095283	88.6	1.082	4.140
18 LN+SwiGLU	17	9.83	2.045564	95.1	1.083	4.146
19 W320 D10	17	9.83	1.872762	490.6	13.029	49.728
20 w384 d8	17	9.83	1.829156	459.7	15.047	57.422

Corresponding run directory (located at code/runs/)

13 recipe_exact_baseline_s17
14 recipe_norepl_s17
15 recipe_norepl_beta095_s17
16 recipe_full_student_s17
17 ablation_rms_gelu_s17
18 ablation_ln_swiglu_s17
19 capacity_w320d10_s17
20 capacity_w384d8_s17

A.3 Location, shared weights, and additional training objectives

Table 29. Retained run indicators

Numbering and setup	Seed	Targets M	Val. BPB	Training seconds	Parameter M	Assets MiB
21 Learned untied	17	9.83	1.786678	392.8	11.226	42.846
22 RoPE tied / seed 17	17	9.83	1.754858	488.1	10.489	40.033
23 RoPE tied / seed 137	137	9.83	1.741184	399.6	10.489	40.033
24 RoPE untied / seed 17	17	9.83	1.745039	473.9	11.144	42.533
25 RoPE untied / seed 137	137	9.83	1.737502	404.5	11.144	42.533
26 RoPE untied + EMA .99	17	9.83	1.734143	363.9	11.144	42.533
27 RoPE untied + MTP k1	17	9.83	1.739964	420.0	11.144	42.533

Corresponding run directory (located at code/runs/)

21 ablation_untied_s17
22 ablation_rope_s17
23 ablation_rope_s137
24 ablation_rope_untied_s17
25 ablation_rope_untied_s137
26 ablation_rope_untied_ema099_s17
27 ablation_rope_untied_mtp1_s17

A.4 Long training and optimal weight selection

Table 30. Retained run indicators

Numbering and setup	Seed	Targets M	Val. BPB	Training seconds	Parameter M	Assets MiB
28 vl long recipe / seed 17	17	28.91	1.622441	1143.1	10.571	40.345
29 vl long recipe / seed 137	137	28.91	1.631858	1092.9	10.571	40.345
30 initial RoPE long training	17	28.91	1.634776	1348.5	10.489	40.033
31 RoPE best / seed 17	17	28.91	1.600916	1435.9	10.489	40.033
32 RoPE best / seed 137	137	28.91	1.588095	1399.2	10.489	40.033
33 RoPE best / seed 233	233	28.91	1.611337	1348.9	10.489	40.033

Corresponding run directory (located at code/runs/)

28 a_e8_lr1e3
29 2026-09-23_seed137_final_recipe
30 2026-09-25_rope_long_seed17
31 2026-09-25_rope_long_best_seed17
32 2026-09-25_rope_long_best_seed137
33 2026-09-25_rope_long_best_seed233

The three RoPE best runs each process 28.909568M targets, with the selected step 2400 weights corresponding to 19.660800M targets. Intermediate weights are better without reducing the cost of training searches. In the initial RoPE long training, only scores are retained and no weights are retained, so they are listed separately and not merged with new runs.

Appendix B: Translucent layered curve chart

The following are curve materials kept in a centralized collection, not additional experiments. B1 - B2 split the six learning rate curves in the main text into two groups, with the remaining figures for easy use. Depth is only distinguished by series, vertical axis truncated, and filling is not an error band. The last point represents the last observation and does not indicate the optimal value. For precise comparison, please refer to the datasheet or the 2D native chart in the PPT.

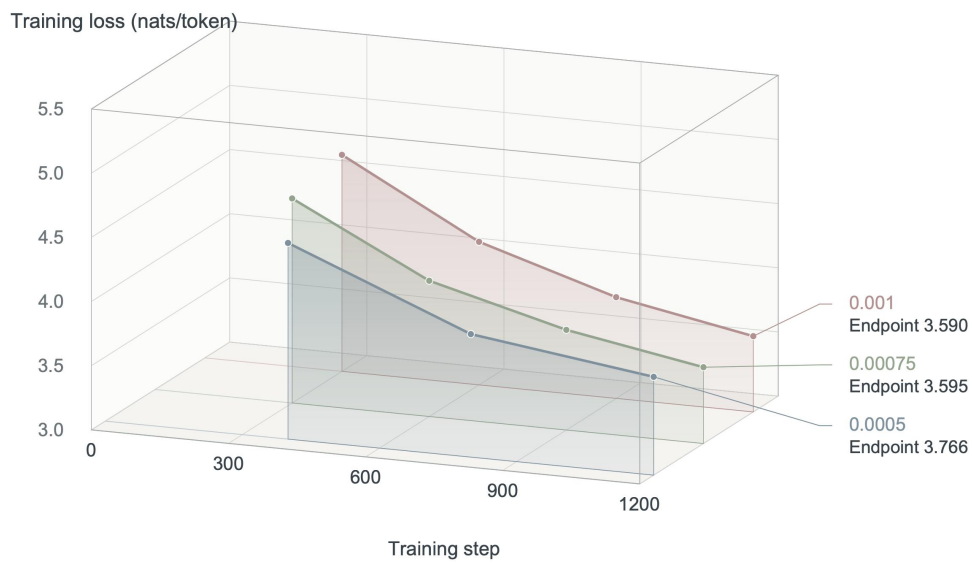


Fig. 20. Training curve A of the learning rate scan. Shares coordinate range with graph B. 0.0005 records points 400, 800, 1200; the others 300, 600, 900, 1200

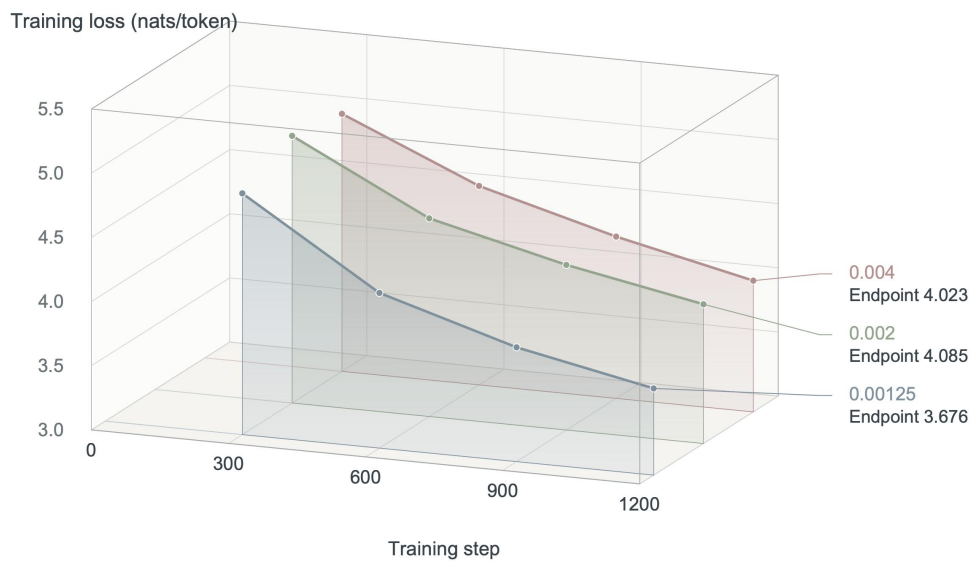


Fig. 21. Training curve B of the learning rate scan. Shares coordinate range with graph A. Training loss is used to check the training process, and learning rate is selected based on the full validation BPB

B.1 v1 and the Initial Long RoPE Run

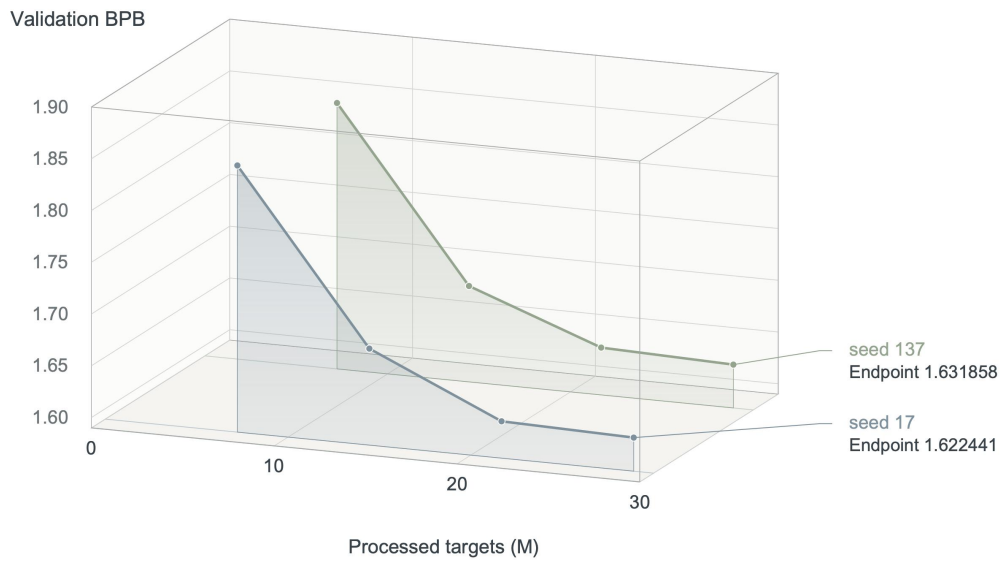


Fig. 22. Two-seed validation curves of the frozen v1 recipe. Each dot comes from the saved complete validation result. Both seeds were trained to 28.91M targets, with no 58M run

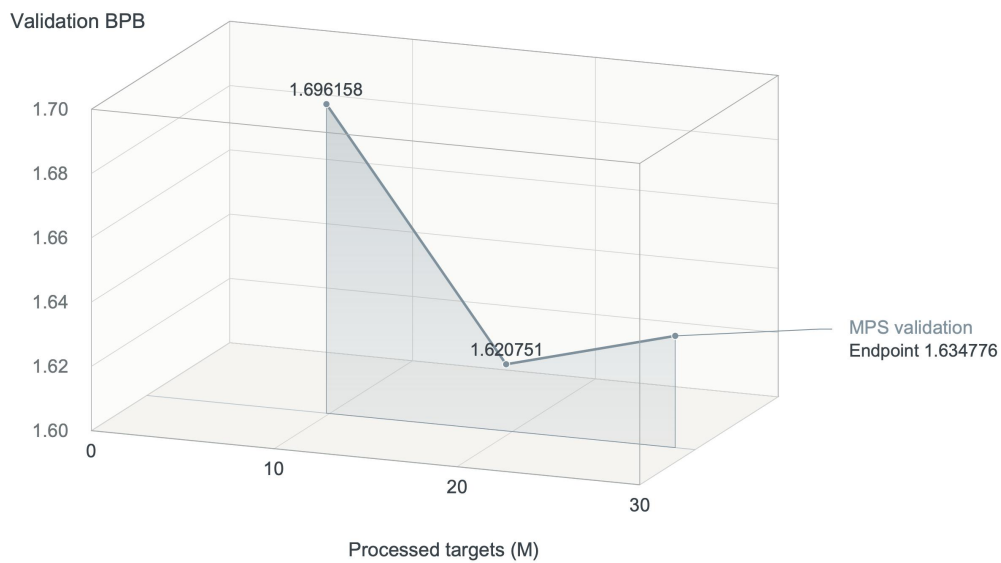


Fig. 23. Validation regression of the initial long RoPE run. The intermediate validation value was low, but the corresponding weights were not retained at that time. This result prompted subsequent runs to save the best checkpoint for pre-declared validation points

B.2 Latest RoPE three-seed system

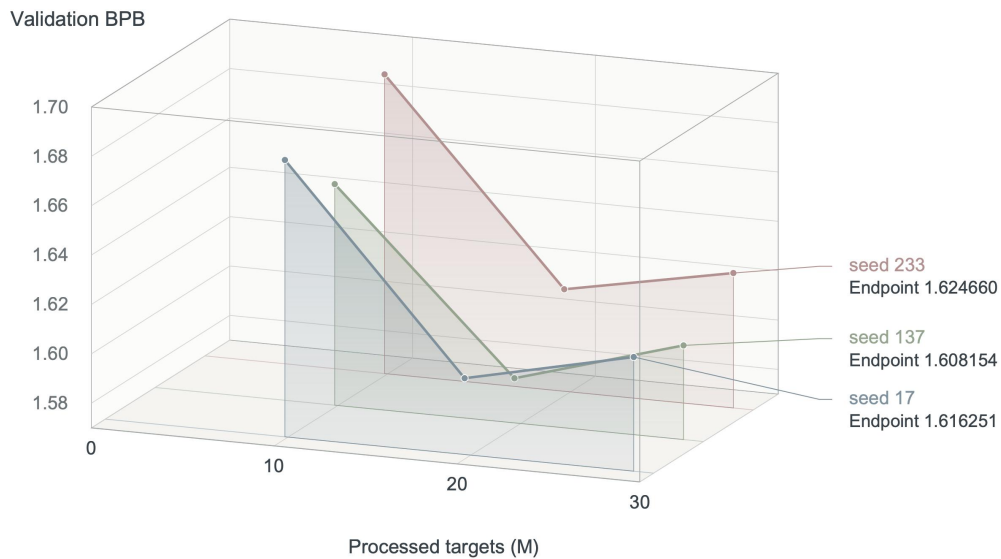


Fig. 24. Validation curves of the three RoPE seeds. All three seeds have weights selected at 19.66M targets. The figure shows MPS validation during training; CPU FP32 verification values are listed separately

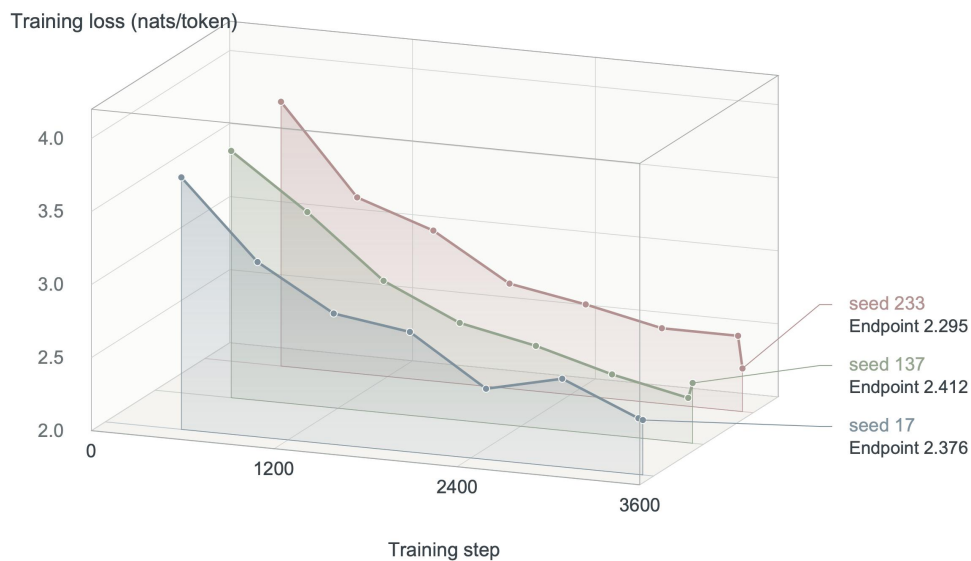


Fig. 25. Training loss of the three RoPE seeds. The dot is the average token NLL of the batch of logs retained, not the complete curve for each step. The connecting line only connects adjacent observations without smoothing or completing data

Appendix C: Manual Modifications and Source of Charts

Main text, tables, and formulas can be edited in Word. Experimental diagrams are uniformly saved in "MP1_Morandi Chart_Editable.pptx": pages 1 - 23 allow modification of walls, curves, columns, transparency, and text; Pages 24 - 40 are 2D native charts with embedded workbooks, which can be right-clicked to "Edit Data". The two versions do not automatically sync. The merged flowchart in Figure 2 remains in a standalone PPT.

Table 31. Word figure index against the Morandi PowerPoint pages

Word drawing number	Experimental content	Styles / Data page
Figure 3	Learning rate validation	3 / 26
Figure 4	RoPE three-seed validation	15 / 38
E1	The main optimization phase	1 / 24
E2	Learning rate versus training loss	2 / 25
E3	Validation results of regular settings	4 / 27
E4	Dual-seed validation curve for the frozen v1 formula	5 / 28
E5	Freeze the official test results for v1	6 / 29
E6	Resource usage and course limits	7 / 30
E7	Trainer settings for comparison	8 / 31
E8	Norm ablation with FFN in 2×2	9 / 32
E9	Short-budget comparisons for deeper or wider models	10 / 33
E10	Location indicates sharing weights	11 / 34
E11	Dual-seed increments untied under RoPE	12 / 35
E12	Additional gains from EMA versus MTP	13 / 36
E13	Late Retrograde After the First RoPE Long Training	14 / 37
E14	Training loss of the three RoPE seeds	17 / 40
E15	v1 paired against the long RoPE recipe	16 / 39

- B6 corresponds to PPT pages 18 - 23. Word illustrations are previews; after modifying the PPT, export the image, then "Change Picture" in Word. The wall-type version is not an Excel linked chart: if the values change, rerun the drawing code to the calculation position; do not drag vertices to represent new results

Data source: `output/figures/full_report/evidence.json`. SVG, color scheme, and page number index: `output/figures/morandi/`. Drawing and Word update code are `code/scripts/build_morandi_report_charts.mjs` and `update_morandi_report_docx.py`, respectively. Please save manual revisions before rebuilding.

After manual modifications, please save as is; rerunning the script will overwrite the generated content. When submitting formally, you may delete editing notes and alternative images, merge duplicate explanations, but you must retain the mechanism reference, resource and frozen evidence, AI disclosure, and meet the course page count requirements.